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# What Does a Benign-Trained Anomaly Detector Actually Detect?

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## Abstract

Anomaly detectors for insider threats are usually judged by how well their scores rank malicious activity above normal activity. We show that this number can be misleading, because it does not reveal what information the detector uses. We build a detector by adapting a large language model to normal employee activity only, so that unusual activity receives a high surprisal score, and we then open the model to determine what its score is based on: we isolate the internal changes that adaptation caused, approximate them with a sparse dictionary, test which dictionary directions causally influence the score, and measure what part of the input those directions respond to. On one standard benchmark the evidence the audit selects turns out to be almost entirely a description of who each user is (their recorded personality and organizational profile) rather than of what they did; on a second benchmark it is mostly behavior. When malicious users are compared only against normal users that the model has never seen, apparent ranking quality falls from about 0.95 to near chance on the first benchmark. The finding survives an extensive robustness program: held-out users, repeated dictionary training, dictionaries trained without any malicious example, a smaller model retrained from scratch with fresh random seeds and a repaired data pipeline, and interventions whose strength is matched between treatment and control. The same analysis on a third dataset, collected from real people with real personality questionnaires, confirms the behavioral case, and classical statistical detectors trained on the same information do not take the profile shortcut. The conclusion is that high ranking performance alone does not show that a detector has learned to detect behavior, and this paper provides a way to check. A train-matched retraining experiment then shows that, in the audited eight-billion-parameter run, profile content in the serialization causes the collapse: removing the profile block from the text used for adaptation and scoring restores unseen-user ranking from chance to about 0.90 (a descriptive paired resampling interval over the four malicious users excludes zero), while the same removal has a small, non-significant effect in the audited three-billion-parameter run; the two runs differ in model family as well as size and were trained once each, so the design does not attribute the difference to capacity alone. The personality line carries the effect. On a corporate login log with no profile of any kind, the collapse on unseen users recurs and host identity, a user-associated field, carries the shortcut.

## 1 The problem

Organizations record what employees do on their computers: logons, file transfers, web activity, e-mail volume, and so on. An *insider threat* is an employee who misuses legitimate access, for example by stealing data. Because such events are rare, a common approach is *anomaly detection*: build a statistical model of normal activity, and flag days that the model finds unusual.

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Formally, a detector is a function  $s$  that assigns a real-valued *anomaly score*  $s(x)$  to each observation  $x$ , where in our setting  $x$  is a record of one user’s activity on one day. Higher scores are meant to indicate more suspicious activity. Detectors are usually evaluated by how well their scores *rank* malicious days above benign ones. The standard summary of ranking quality is the following probability, estimated over a labeled test set:

$$\text{AUC} = \Pr(s(X_{\text{mal}}) > s(X_{\text{ben}})), \quad (1)$$

where  $X_{\text{mal}}$  is a randomly chosen malicious day and  $X_{\text{ben}}$  a randomly chosen benign day. (This quantity is commonly called the ROC-AUC, for “area under the receiver operating characteristic curve”; the probability formulation is equivalent and is the only form we need.) An AUC of 1 means perfect ranking and 0.5 means the ranking is no better than chance.

The question that motivates our work is simple to state. A high AUC says that the score *separates* the two classes. It does not say *on the basis of what information* the separation is achieved. Two detectors with identical AUC could work very differently: one might respond to genuinely unusual behavior, while the other might respond to incidental properties of *who* the malicious users happen to be, such as their role, department, or personality profile. The second kind of detector would be nearly useless in practice, because in deployment the malicious users are not known in advance. This paper develops a method for determining which kind of detector one actually has, applies it to a modern detector built from a large language model, and then subjects the answer to a systematic robustness program.

## 2 The data

We use two versions (called r4.2 and r6.2) of the CERT insider-threat corpus, a widely used synthetic dataset that simulates a corporation’s activity logs, produced by Carnegie Mellon University. Each version contains several thousand simulated employees observed over about eighteen months, with a small set of employees scripted to carry out malicious scenarios on particular days. Version r4.2 has sixty malicious users among about one thousand; version r6.2 has four among about four thousand. This difference in composition turns out to matter, and we return to it repeatedly. A third dataset, TWOS, collected from real people, is described in Section 7.

Each user-day is converted into a short structured text with three kinds of lines:

- a **DAY** line holding static organizational fields for that user (project, role, business unit, department, team, and an information-technology administrator flag);
- a **PSY** line holding the user’s five simulated personality scores. These are the “Big Five” personality traits used in psychology: openness (O), conscientiousness (C), extraversion (E), agreeableness (A), and neuroticism (N). In CERT these numbers are generated by the simulator and are constant for each user;
- one **SES** line per work session, holding numeric behavioral counts for that session: duration, number of logons, file and removable-drive activity, e-mail volume, web-browsing category counts, and similar quantities.

A user-day text is therefore a small table serialized as text. The first two lines describe *who the user is* and are essentially constant across a user’s days; the remaining lines describe *what the user did* that day. We call the first two line classes *profile* content and the third *behavioral* content. This partition of the input is the reference frame for the entire paper.

Users are split as follows. Users with any malicious day are reserved entirely for evaluation. The remaining benign users are divided into a training set (roughly ninety percent) and a validation set (roughly ten percent) whose days never enter training. The validation users will play an important role: they are the only benign users about whom the trained model knows nothing.

### 3 The detector

#### 3.1 Language models as probability distributions

A *language model* is a parametric probability distribution over sequences of symbols called tokens. Writing a sequence as  $x = (x_1, \dots, x_n)$ , the model has the product form

$$p_\theta(x) = \prod_{t=1}^n p_\theta(x_t \mid x_1, \dots, x_{t-1}), \quad (2)$$

where each conditional distribution is computed by a neural network with parameters  $\theta$ . Modern large language models are such networks with billions of parameters, trained on enormous general text corpora. We use one called Qwen3-8B (eight billion parameters). The network processes a sequence through  $L$  successive layers; at layer  $\ell$  and position  $t$  it maintains an internal state vector  $h_t^{(\ell)} \in \mathbb{R}^d$  with  $d = 4096$ . These internal vectors matter later: they are the objects we analyze.

#### 3.2 Adapting the model to normal behavior only

We specialize the general-purpose model to our corpus by *fine-tuning*: continuing its training on the serialized user-day texts, using only days from benign training users. The objective is maximum likelihood, that is, minimizing the negative log-likelihood of the benign corpus  $\mathcal{D}$ :

$$\mathcal{L}(\theta) = - \sum_{x \in \mathcal{D}} \sum_t \log p_\theta(x_t \mid x_{<t}). \quad (3)$$

No malicious example is ever used in training. This is the classical one-class setup: model the normal population, then measure deviation from it.

For efficiency the fine-tuning does not modify all eight billion parameters. Each weight matrix  $W$  of the network is kept fixed and receives a trainable low-rank correction,

$$W' = W + \frac{\alpha}{r} BA, \quad B \in \mathbb{R}^{d_{\text{out}} \times r}, A \in \mathbb{R}^{r \times d_{\text{in}}}, \quad (4)$$

with rank  $r = 16$ , so only the small matrices  $A$  and  $B$  are learned. This technique is called LoRA (low-rank adaptation); we use a memory-saving variant called QLoRA in which the fixed weights are stored at reduced numerical precision. The pair (fixed base model, learned low-rank correction) is called the *adapted model*. The important point is that the entire effect of training is carried by a low-dimensional, removable perturbation, which makes before-versus-after comparisons clean.

#### 3.3 The anomaly score and two ways to evaluate it

The score of a day  $x$  is its average surprisal under the adapted model:

$$s(x) = -\frac{1}{n} \sum_{t=1}^n \log p_{\theta_{\text{ada}}}(x_t \mid x_{<t}). \quad (5)$$

Days that the model of normal behavior finds improbable receive high scores.

There are two reasonable ways to estimate the AUC of equation (1), and the difference between them will turn out to be one of the paper’s findings.

1. *The standard protocol.* Malicious days are compared against benign days drawn from the general population. Because the model was trained on about ninety percent of benign users, most of those benign comparison days belong to users the model saw during training. Under this protocol the adapted model looks excellent: AUC about 0.953 on r6.2 and 0.964 on r4.2, higher than four standard anomaly-detection baselines evaluated on the same benchmarks (an isolation forest, a deep one-class method called Deep SVDD, and two recurrent autoencoders), whose AUC values range from about 0.63 to 0.77.

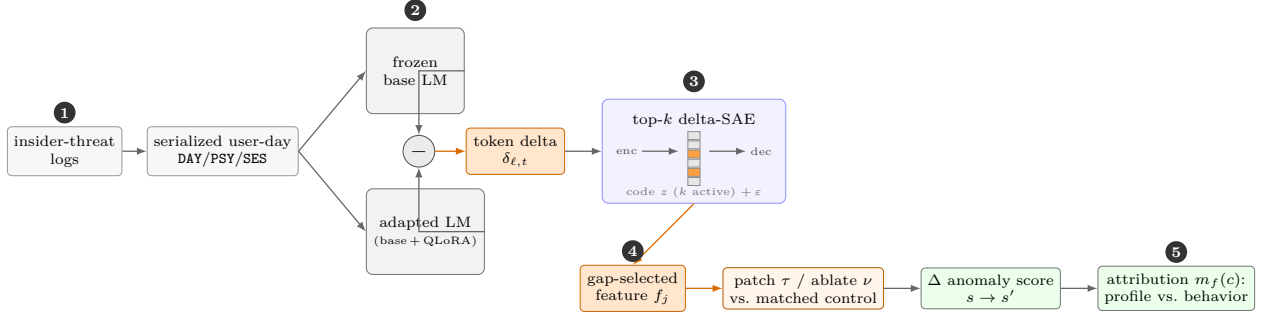


Figure 1: The delta-SAE audit, drawn as architecture. **(1)** A user-day is serialized into structured DAY/PSY/SES text. **(2)** The same input runs through the frozen base model and its benign-only QLoRA-adapted copy; at layer  $\ell$  we take the per-token residual-stream difference  $\delta_{\ell,t} = h_{\ell,t}^{\text{ada}} - h_{\ell,t}^{\text{base}}$  (the adapter is left abstract—the delta is the object of study). **(3)**  $\delta$  feeds a top- $k$  sparse autoencoder (encoder  $\rightarrow$  sparse code  $z$  with only  $k$  units active, shown as the lit cells  $\rightarrow$  decoder, plus reconstruction error), whose decoder columns form the feature dictionary. **(4)** A gap-selected feature  $f_j$  is causally patched ( $\tau$ ) or ablated ( $\nu$ ) against a matched active control, and the effect on the adapted anomaly score  $s \rightarrow s'$  is measured. **(5)** Token attribution  $m_f(c)$  labels the feature’s content as profile- or behavior-bound. A mechanistic claim requires both causal relevance (Step 4) *and* content (Step 5); formal definitions are in the main paper.

2. *The user-disjoint protocol.* Malicious days are compared only against benign days of the validation users, whom the model never saw. Malicious users are unseen by construction, so under this protocol both sides of the comparison are equally unfamiliar to the model.

The gap between these two numbers measures how much of the apparent performance is explained by mere familiarity with specific users rather than by properties of the activity itself. We report the gap in Section 5.2.

## 4 Opening the model

We now describe the four analysis steps, in the order they are applied. Together they take us from the raw network to a statement about what the score is based on.

Figure 1 draws the whole chain as an architecture before we walk through it step by step.

### 4.1 Step 1: isolate what fine-tuning changed

Because the base model is kept frozen, we can run the same day  $x$  through both the base and adapted models and subtract their internal states. Define, at a chosen layer  $\ell$  and each token position  $t$ ,

$$\delta_t = h_t^{(\ell),\text{ada}} - h_t^{(\ell),\text{base}} \in \mathbb{R}^d. \quad (6)$$

The vector  $\delta_t$  isolates the representational change induced by the adaptation at the audited layer and position. It is the natural object to study if one wants to know what the adaptation, as opposed to the general-purpose model, contributes at that point in the computation.

### 4.2 Step 2: approximate the changes with a sparse dictionary

The vectors  $\delta_t$  live in  $\mathbb{R}^{4096}$  and are not individually interpretable. We therefore approximate them by sparse nonnegative combinations of learned dictionary elements. After standardizing each coordinate (subtracting its mean  $\mu$  and dividing by its standard deviation  $\sigma$ ), we seek a dictionary matrix  $D \in \mathbb{R}^{d \times m}$  (with  $m$  columns, called *features*;  $m = 4d$  for the headline r6.2 audit and  $m = 2d$  for the native r4.2 audit) and, for each standardized vector  $\tilde{\delta}_t$ , a coefficient vector  $z_t \in \mathbb{R}_{\geq 0}^m$  with at most  $k$  nonzero entries ( $k = 8$  and  $k = 4$  respectively), such that

$$\tilde{\delta}_t \approx Dz_t, \quad (7)$$

with  $D$  and the coefficients chosen to minimize the mean squared reconstruction error over a large sample of token positions. This is a sparse coding problem; the particular parametrization we use (a one-layer encoder network followed by a keep-the-largest- $k$  rule and a linear decoder) is called a *sparse autoencoder* in the machine-learning literature. The outcome is that each change vector is expressed as a short combination of reusable directions, and each direction can be studied on its own. One technical point deserves notice: because the standardization constants differ between datasets, any geometric comparison of features across datasets must first map the dictionary columns back to the original coordinates, that is, compare the directions  $\sigma \odot D_{:,f}$  rather than the raw columns. All cross-dataset comparisons in this paper do so.

### 4.3 Step 3: select the score-relevant features and test them causally

Some dictionary features activate similarly on malicious and benign days; those cannot carry the anomaly signal. We rank features by the difference in mean activation,

$$g_f = \mathbb{E}[z_{t,f} \mid \text{malicious day}] - \mathbb{E}[z_{t,f} \mid \text{benign day}], \quad (8)$$

and take the five features with the largest  $g_f$  as the candidate set  $S$ . As a comparison group we also select a matched *control* set  $C$  of five features that are similarly active overall but carry no class difference (small  $|g_f|$ ). Selection by a statistic can be flattered by chance, so everything that follows treats  $S$  as a hypothesis to be tested rather than a conclusion.

The test is interventional. The idea is to edit the coefficients of the selected features inside the model, let the edited internal state flow through the remaining layers, and observe whether the anomaly score responds. We use two complementary edits.

*Repair (a sufficiency-style test).* For a malicious day, we move the selected coefficients toward the values they take on a comparable benign day (a “donor” from the same organizational context, never the same user), with an interpolation strength  $\alpha \in \{0.25, 0.5, 0.75, 1\}$ , and record the best achievable reduction of the score over the available donors and strengths. Write  $\Delta_{\text{ben}}^S$  for this quantity, and  $\Delta_{\text{anom}}^S$  for the analogous quantity when donors are other malicious days. If the selected features carry anomaly-specific repairable structure, moving them toward benign values should reduce the score more than moving them toward other anomalous values, and it should do so more for  $S$  than for the control set  $C$ . The reported estimand is the doubly differenced quantity

$$\tau = (\overline{\Delta_{\text{anom}}^S} - \overline{\Delta_{\text{ben}}^S}) - (\overline{\Delta_{\text{anom}}^C} - \overline{\Delta_{\text{ben}}^C}). \quad (9)$$

The double difference protects the comparison: an effect that comes merely from making an edit of that size, rather than from the specific content of the selected features, appears in both arms and is removed to the extent that the control edit matches the treatment edit in strength, which the paper enforces by matching the size of the control edit to the treatment edit.

*Ablation (a necessity-style test).* We scale the selected coefficients toward zero, on malicious days and on matched benign days, and ask whether removal disrupts the score more on the malicious side and more for  $S$  than for  $C$ . The corresponding double difference is denoted  $\nu$ .

Two safeguards are essential and were arrived at the hard way. First, donors from the same user are excluded: allowing them lets the procedure “repair” a malicious day by borrowing the same person’s own benign signature, which inflates effects for reasons unrelated to the question. Second, uncertainty is computed by resampling *users*, not days: multiple days of the same user are statistically dependent, so all confidence intervals in this paper are user-level bootstrap intervals with the malicious user as the resampling unit (10,000 draws). On r6.2, which has only four malicious users, such intervals are reported as descriptive.

One further check matters because the selected features are, by construction, strongly activated: their edits are physically larger than edits of the control features. We therefore repeated the headline comparisons with *dose-matched* controls, chosen to match the selected features on activation frequency, activation magnitude, and dictionary-direction norm, which closes the edit-size gap to a factor of 1.3 to 1.8. The contrasts survive: on r4.2 the repair contrast remains clearly positive (+0.0017 with interval [0.0011, 0.0023]) and on r6.2 the ablation contrast remains clearly positive (+0.061 with interval [0.007, 0.084]); the remaining two contrasts stay positive but with intervals containing zero. Figure 2 summarizes the intervention measurements.

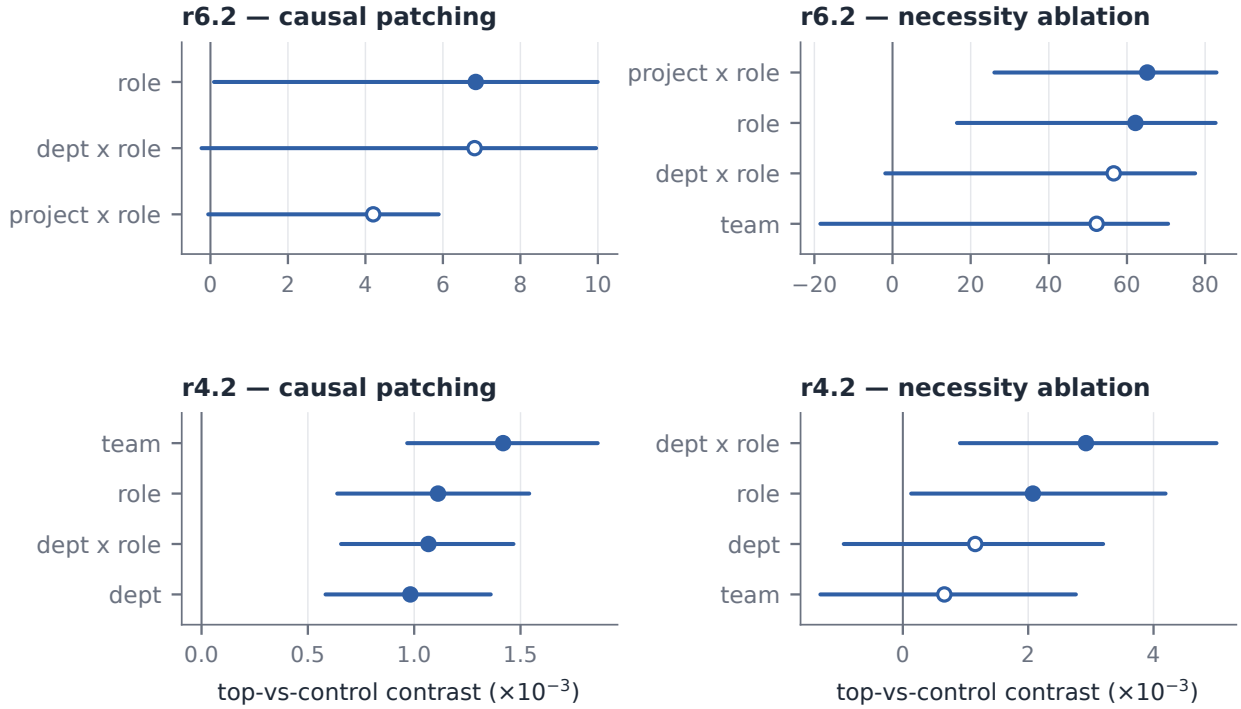


Figure 2: Effect of editing the selected features, compared with editing matched control features, on the anomaly score (one panel per benchmark and per type of edit; units are thousandths of the score). Each point is the average effect for one organizational matching context, and each horizontal bar is a 95 percent confidence interval computed with the malicious users as the resampling unit. Positive values mean that the selected features influence the score more strongly than the control features do. Open circles mark intervals that include zero.

#### 4.4 Step 4: read off what the features respond to

Finally we ask *where* the selected features activate within the day text. Every token position belongs to one line class: the organizational **DAY** line, the personality **PSY** line, or a behavioral **SES** line. For a feature  $f$ , define its *attribution mass* on class  $c$  as the fraction of its total activation that occurs at positions of that class:

$$m_f(c) = \frac{\sum_{(x,t) : \text{class}(t)=c} z_{t,f}(x)}{\sum_{(x,t)} z_{t,f}(x)}. \quad (10)$$

If a feature has nearly all of its mass on **PSY** and **DAY** positions, it responds to who the user is; if its mass lies on **SES** positions, it responds to what the user did.

The logic of the whole method is a conjunction. The interventions of Step 3 establish that the selected features are causally connected to the score, but say nothing about their meaning. The attribution of Step 4 establishes what the features respond to, but says nothing about whether they matter. Only together do the two steps answer the question in the title, and the paper’s findings repeatedly illustrate that either step alone would have been misleading.

## 5 Findings on the two CERT benchmarks

### 5.1 Two detectors that look alike and are not

The two CERT versions produce nearly identical headline AUC values, yet the analysis reveals qualitatively different detectors.

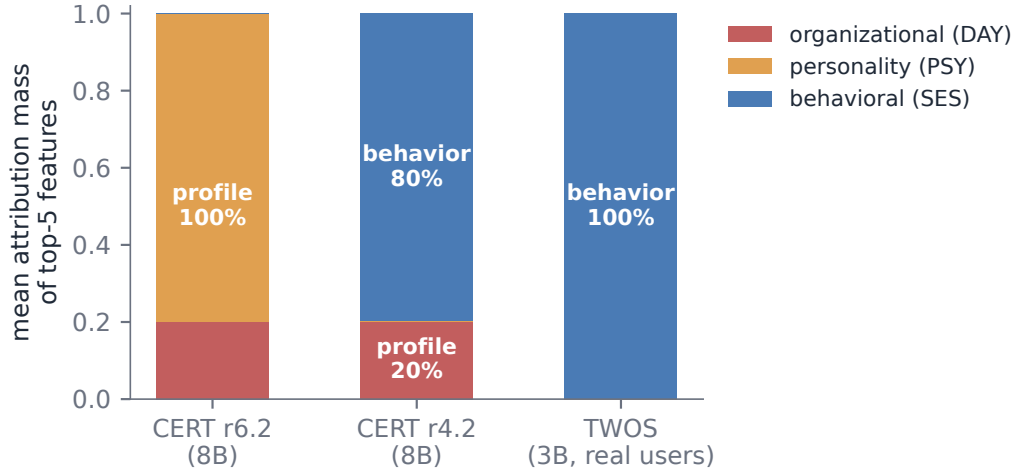


Figure 3: Where the selected features place their activation, averaged over each detector’s five selected features. Each bar decomposes the activation mass into the organizational lines, the personality lines, and the behavioral session lines of the input text. The r6.2 detector’s selected features respond almost entirely to who the user is; the r4.2 and TWOS detectors’ selected features respond almost entirely to what the user did.

On r6.2, the five selected features place between 99.8 and 100 percent of their attribution mass on the personality and organizational lines, with activation on the personality line enriched about thirteen times over its share of tokens. The intervention contrasts are positive (repair  $\tau = 0.0068$  with descriptive interval  $[0.0001, 0.0100]$ ; ablation  $\nu = 0.065$  with interval  $[0.026, 0.083]$ ). The evidence the audit selects, the five features that most raise the score, is almost entirely a description of *who the user is*.

On r4.2, four of the five selected features concentrate on the behavioral session lines (the fifth responds to the organizational header). The contrasts are again positive (repair  $\tau = 0.0014$  with interval  $[0.0010, 0.0019]$ ; ablation  $\nu = 0.0029$  with interval  $[0.0009, 0.0050]$ ). There, the evidence is mostly *what the user did*. Figure 3 shows the contrast, together with the real-data result of Section 7.

## 5.2 The consequence for the headline numbers

Why would reading a user’s static profile produce a high AUC at all? The adapted model was trained on roughly ninety percent of the benign users. For those users, the model has been repeatedly exposed to the constant profile lines, so their profile text is comparatively familiar and their days look unsurprising. The malicious users were never seen in training, so their profile lines alone make their days look surprising. The score is then partly measuring *familiarity* rather than behavior.

The user-disjoint protocol of Section 3.3 removes this asymmetry, and the effect is large. The AUC falls from 0.954 to 0.545 on r6.2, which is barely better than chance, and from 0.959 to 0.668 on r4.2. Under the fair comparison the language model no longer outperforms the classical baselines at all. Two further measurements complete the picture. First, ranking each malicious user’s own days (their malicious days against their own benign days, identity held fixed) still succeeds moderately on both benchmarks (average within-user AUC 0.67 on r4.2 and 0.73 on r6.2), so behavioral signal exists; it is simply drowned, in the cross-user comparison, by the familiarity effect. Second, deleting the personality lines from the inputs at scoring time *improves* the r4.2 detector’s performance on unseen users from 0.67 to about 0.80 (an increase of 0.127 with interval  $[0.061, 0.193]$ ): the profile text was not merely unhelpful, it was actively obscuring the behavioral signal. On r6.2, deleting profile content changes nothing that is statistically distinguishable; with every user unseen, profile novelty separates nothing, and no clear behavioral signal is revealed underneath. Figure 4 shows the baseline comparison and the collapse.

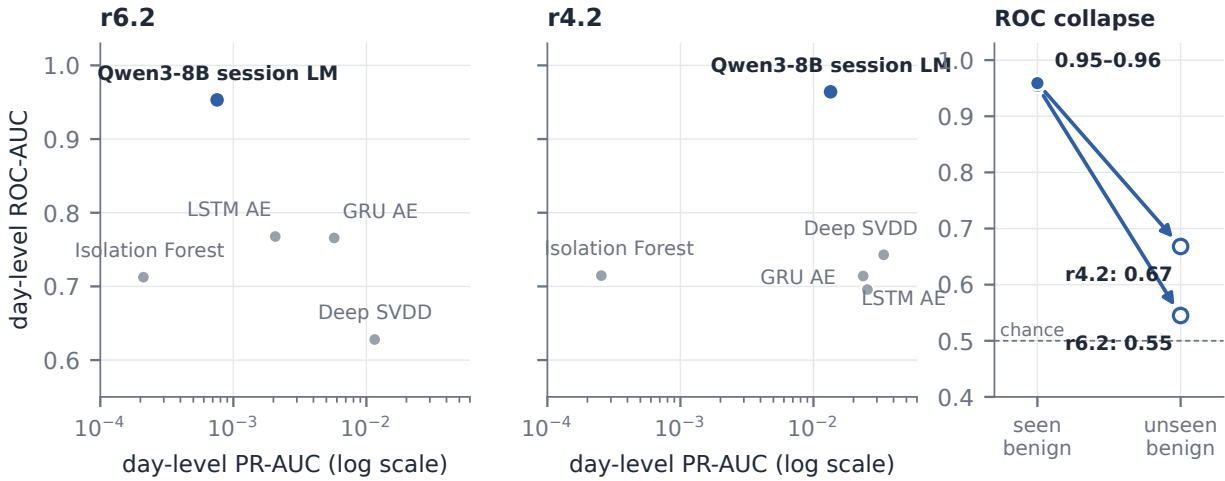


Figure 4: Score-level results. Left and center panels: ranking quality (vertical axis) and precision (horizontal axis, logarithmic scale) for the language-model detector (blue) and four standard anomaly-detection methods on the two benchmarks; the language model attains the highest ranking quality under the standard protocol. Right panel: the same language-model ranking quality when malicious users are compared only against benign users never seen during training. The apparent advantage largely disappears, and the dashed line marks chance level.

## 6 Is the interpretation robust?

A single analysis pipeline contains many arbitrary choices, and each is an opportunity for an artifact. This section walks through the checks, one question at a time. Each check either survived or, where it did not, the failure is reported and incorporated into the claims.

*Does the finding survive on users that played no role in selecting the features?* On r4.2 we split the sixty malicious users into two halves, re-selected features using only the first thirty, and estimated effects only on the second thirty. Four of the five selected features reappear, and all effect estimates remain positive on the held-out half, at roughly half the original magnitude, with one matching context individually excluding zero. The shrinkage is the expected signature of selection effects, and the claims are scoped accordingly. On r6.2, with only four malicious users, we used leave-one-user-out selection instead; there the effects concentrate on the single dominant user, who contributes 46 of the 70 malicious days, and the paper reports r6.2 causal claims at that resolution.

*Is the dictionary itself an accident of its random initialization?* Training the dictionary three times with different random seeds and comparing the resulting feature directions (in original coordinates, as noted in Step 2) shows that each benchmark’s selected feature family is stable: matched features across seeds align with cosine similarity 0.81 to 0.95, above a null baseline of 0.72 to 0.82 computed from the full dictionaries. Across benchmarks, by contrast, the selected features align at or below the null; notably, where the two dictionaries share the most overall geometry, the selected features align *worse* than typical features. The two detectors’ evidence is not the same structure under different labels.

*Does the finding depend on malicious examples having shaped the dictionary?* The dictionaries above were fitted on a pool that includes malicious days. We retrained both dictionaries from scratch on benign rows only, so that no malicious example influences dictionary formation, and repeated selection, attribution, and evaluation. The attribution dissociation reproduces exactly (r4.2 selected features at least 94 percent behavioral; r6.2 features at least 99 percent profile-bound in every leave-one-user-out fold). The intervention results are estimand-dependent under the new dictionaries: the ablation contrast on r4.2 replicates across all matching contexts, while the repair contrast does not. We therefore treat ablation dependence as the robust causal signature and report repair effects with that qualification.

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*Is the difference between the benchmarks simply that one has four malicious users and the other sixty?* We subsampled r4.2’s malicious population down to four users (ten random repetitions per size) and re-ranked features each time. The profile share of the selected features stays low and nearly flat across population sizes. Raw positive count alone does not explain the dissociation; the composition of the population, not its size, is what matters.

*Does the finding survive a fresh model, fresh randomness, and a repaired pipeline?* A data-serialization defect (four file-transfer fields silently dropped) was discovered during the project. We repaired it and reran the entire pipeline at smaller scale: a different, smaller language model (Qwen2.5-3B) trained from scratch twice per benchmark with different random seeds, new dictionaries trained on benign rows only, and fresh selection and attribution. The dissociation reproduces under both seeds: r4.2 selected features are behavioral in every audited configuration, and r6.2 develops features that are 100 percent profile-bound at the deepest audited layer under both seeds. The ablation contrast again replicates on r4.2 across seeds (+0.0022 and +0.0021) while the repair contrast again does not, matching the pattern above.

*Could the score’s profile sensitivity be an illusion of the analysis rather than a property of the score?* Two input-level experiments require no interpretability machinery at all and agree with the internal analysis. Replacing a malicious day’s profile lines with a random benign user’s profile changes the score more than replacing its entire behavioral content on both benchmarks (mean score change +0.155 against +0.072 on r6.2, a ratio of 2.2; +0.180 against +0.102 on r4.2, a ratio of 1.8). The profile swap dominates on both; the behavioral swap is a larger share of the total on r4.2, in line with the more behavioral audit there, but the two benchmarks are not reversed. And the masking result of Section 5.2 shows the score-level consequence directly.

## 7 Real data with real personality scores

The CERT data are simulated, including the personality scores. The natural objection is that the entire phenomenon might be an artifact of simulation. We therefore repeated the pipeline on TWOS, a public dataset collected at the Singapore University of Technology and Design from twenty-four real users over one week, during which, by design of the underlying study, twelve users had their sessions temporarily taken over by other participants (a “masquerade” attack), and five users, after a staged dismissal, briefly acted as insiders on their own accounts (a “traitor” scenario). Each real user completed the standard fifty-item Big Five questionnaire, so the serialized inputs contain genuine psychometric information.

The details mirror the CERT setup at smaller scale. Mouse, keyboard, and logon telemetry are aggregated into ten-minute windows and serialized in the same three-line-class format, with the real Big Five scores forming the personality line (18,048 windows in total, of which 138 overlap a staged attack: 101 masquerade and 37 traitor windows). The model is the smaller Qwen2.5-3B, adapted on benign windows only (about seven thousand windows from the twelve users who were never masqueraded; for the five later dismissed users, only days before the dismissal enter training), trained twice with different random seeds; dictionaries are trained on benign rows only; selection and attribution proceed exactly as before.

Because half of the users are attacked, profile information is a far weaker separator here than in CERT r6.2, and our account predicts that the detector should rely on behavior. That is what the analysis finds. The selected features place essentially all of their attribution mass on the behavioral lines, under both training seeds and at both audited layers (the smallest behavioral share of any selected feature is 0.983). Decisively, the evaluation compares each attacked user’s attack windows with that same user’s normal windows, so static identity and personality information is held fixed and cannot explain the observed separation; the average within-user AUC is 0.78 across the sixteen attacked users (the twelve masqueraded users plus the four dismissed users with logged activity in their staged window). The intervention experiments reproduce the pattern seen on CERT: the ablation contrast is positive (+0.0019) while the repair contrast is not, the fourth independent occurrence of that asymmetry. The behavioral pole of the dissociation is not an artifact of synthetic data or simulated psychometrics.

## 8 Real data with no profile at all

TWOS still contains a profile line, albeit a real one, so a determined skeptic can maintain that the shortcut needs *some* injected profile to exist. The two-factor account makes a sharper prediction than that: wherever the serialized text carries a field that is constant for each user and reveals who the user is, a likelihood-based detector will shortcut on *that* field, whatever it is called. We tested this on the “Comprehensive, Multi-Source Cyber-Security Events” dataset released by Los Alamos National Laboratory in 2015: fifty-eight days of raw Windows authentication events from a real enterprise (12,425 users, 1.05 billion events), with 749 red-team compromise events (credential-based lateral movement) over 104 users as ground truth and no profile of any kind. Each event records only identity fields (source and destination user, source and destination host) and behavioral fields (authentication type, logon type, orientation, outcome, hour).

We serialized each user-day into windows of thirty-two consecutive events, one text line per event, and called a window positive if it contained a red-team event; the longest window is 1,503 tokens against a 2,048-token context, so no window was truncated. We kept every red-team user and a one-in-ten sample of benign users (1.62 million windows, 432 of them positive), adapted Qwen2.5-3B on benign windows only under the same recipe as the CERT experiments, and scored held-out windows by adapted negative log-likelihood. Users were hashed into five folds; the adapters trained on benign windows from four folds (300,729 windows) and were evaluated separately on seen-fold users (22,563 windows, 234 positive) and on the held-out fold (30,410 windows, 198 positive). Four serializations differ only in the identity fields: the full line, user identity anonymized, host identity anonymized, and user identity deranged across users. Positive windows are rare (prevalence 1.0 percent among seen-fold windows and 0.65 percent among unseen-fold windows), so we report both AUC and average precision.

| Identity fields in the serialization | Seen AUC | Seen AP | Unseen AUC   | Unseen AP |
|--------------------------------------|----------|---------|--------------|-----------|
| full (user + host)                   | 0.949    | 0.177   | 0.497        | 0.013     |
| user anonymized                      | 0.944    | 0.176   | 0.518        | 0.019     |
| host anonymized                      | 0.877    | 0.069   | <b>0.627</b> | 0.013     |
| user deranged (shuffle)              | 0.947    | 0.173   | 0.488        | 0.012     |

Three observations. First, the seen-versus-unseen collapse reproduces on real data with no injected profile: every serialization ranks seen users near 0.95 and unseen users far lower, at chance except when hosts are anonymized. Second, the user field contributes little: anonymizing or deranging it moves unseen-user AUC by at most 0.02 (a fixed permutation relabels a user rather than removing the label), so the shortcut is not carried mainly by the user identifier. Third, the host field carries it, with the trade-off seen for CERT’s profile block: anonymizing hosts costs seen-user ranking (0.949  $\rightarrow$  0.877) and improves unseen-user ranking (0.497  $\rightarrow$  0.627), although unseen-user average precision stays at the base rate (0.013 at prevalence 0.65%) in every condition, so the gain is in ranking, not in alert precision. Hosts are user-associated rather than per-user-constant: a user’s most frequent source host accounts for a median 54% of that user’s events, and only 7% of users issue 80% or more of their events from one host. The host field is therefore a partial identity proxy that may also encode resource access and attack-path structure, and its analogy to CERT’s constant personality line is qualitative. A seen-versus-unseen gap remains within the host-anonymized condition (0.877 versus 0.627), which may reflect behavioral heterogeneity across users, residual identity cues, or both. Two caveats: this is a single 3B adapter per condition, and the ground truth is red-team credential misuse rather than CERT’s insider scenarios, so these numbers replicate the *mechanism*, not a detection benchmark. Notably the host shortcut is severe even in this 3B adapter, whereas on CERT r6.2 the profile-removal effect appears only in the audited 8B run (Section 10), consistent with the availability factor being a property of the data.

## 9 Which detectors take the shortcut?

One further question separates two very different readings of the r6.2 result. Is the profile information so predictive in that dataset that *any* statistical method would exploit it, or is the shortcut a property of this particular detector’s construction?

To answer it we trained three classical anomaly detectors on the same r6.2 days, represented as ordinary numeric feature vectors that *include* the profile variables (twelve profile columns among 242 behavioral ones): an isolation forest, a one-class support vector machine, and a principal-component reconstruction method, each trained only on benign training users and evaluated user-disjointly, exactly as the language model was. For attribution we use permutation importance, which is defined without reference to any particular model class: for each input column, randomly permute that column’s values across the evaluation days, recompute the AUC, and record the decrease; the importance of a group of columns is its share of the total decrease.

All three classical detectors essentially ignore the profile variables. The profile columns receive four to seven percent of the total importance, no profile column appears among any detector’s five most important columns, and deleting the profile columns entirely does not reduce any detector’s AUC (it slightly increases it, by 0.003 to 0.022). The same holds on r4.2. The shortcut is therefore not an inevitable consequence of the available profile information: we do not observe comparable profile reliance in the three classical baselines. An explanation consistent with this evidence lies in the training objective. For a likelihood-based sequence model, the per-user-constant profile tokens are the most predictable part of every training sequence, so learning them yields a large and cheap reduction of the training loss; in a plain feature-vector representation the same information is twelve columns among 242, with no comparable incentive attached. This also explains a number from Section 5.2: under the fair protocol the language model *underperforms* the classical baselines on r6.2, exactly as expected if its adaptation spent capacity on a shortcut that does not generalize while the baselines learned behavior.

## 10 Is the profile the cause? A train-matched experiment

Every causal result so far intervenes at scoring time on a detector that was trained with the profile present, so profile capture could in principle be a correlate of adaptation rather than a consequence of the profile. The direct test is a controlled experiment that changes the text used for adaptation and scoring. We trained four adapters from scratch under an identical recipe on r6.2 texts that differ *only* in the profile block: (i) FULL; (ii) NO-PSY, with the personality line removed; (iii) NO-PROFILE, with both profile lines removed; and (iv) SHUFFLE, a seeded derangement that gives every user another user’s profile, so that token presence, format, and surface statistics and per-user constancy are preserved; only the original correspondence between a user and the recorded attributes is broken, and the reassigned profile still acts as a stable pseudonym. We ran this factorial at both scales used in the paper, Qwen2.5-3B and the headline Qwen3-8B (300,000 benign training examples, one adapter seed per scale).

Each adapter was scored under the strictest protocol available, the fold-aligned protocol of Section 3.3: each of r6.2’s four malicious users is held out in turn and ranked against all 406 benign users of this pool never used in training (the protocol’s cap of 800 exceeds the pool), a user’s score is the maximum of its daily scores, and the AUC is averaged over the four folds. Uncertainty is the cluster bootstrap that resamples the four malicious users ten thousand times; because the same users are resampled for every arm, it also yields a paired interval for the difference between two arms. With four malicious users these intervals are descriptive: they are conditional on the trained adapters and the fixed comparison cohort, and they say nothing about variation across training runs.

| Training serialization    | 3B unseen-user AUC   | 8B unseen-user AUC          |
|---------------------------|----------------------|-----------------------------|
| FULL                      | 0.926 [0.84, 0.98]   | 0.532 [0.25, 0.82]          |
| NO-PSY                    | 0.903 [0.87, 0.93]   | <b>0.906</b> [0.88, 0.93]   |
| NO-PROFILE                | 0.956 [0.88, 0.99]   | <b>0.903</b> [0.84, 0.96]   |
| SHUFFLE                   | 0.656 [0.50, 0.81]   | 0.485 [0.29, 0.68]          |
| removal (NO-PROFILE–FULL) | +0.03 [−0.07, +0.14] | <b>+0.37</b> [+0.07, +0.69] |

*In the audited 8B run the collapse is caused by profile content and is reversible.* The 8B FULL adapter collapses on unseen users (user-level AUC 0.532; day-level 0.399), qualitatively reproducing the headline collapse of Section 5.2 on a capped-corpus adapter. Removing the profile restores user-level AUC to 0.903, a paired removal effect of +0.371 whose descriptive bootstrap interval excludes zero (one-sided bootstrap tail

fraction 0.005). The four per-user differences are  $-0.05$ ,  $+0.82$ ,  $+0.42$ , and  $+0.29$ : the effect is carried by three of the four users, and the interval characterizes evaluation uncertainty for these two adapters rather than replacing replication across adapter seeds. Dropping the personality line alone recovers as much (0.906): the psychometric line is the carrier, and the organizational header adds little. Because each condition changes the serialization used for both adaptation and scoring, the result attributes the collapse to profile content in the pipeline rather than to a training-only pathway; within that scope it is causal, not a correlate.

*In the audited 3B run the shortcut is captured but removal has little effect.* The 3B FULL adapter already generalizes (0.926; its day-level AUC of 0.741 is in line with the better of the paper’s two repaired-serializer 3B seeds, 0.738, Section 6), and removal changes little ( $+0.03$ , interval spanning zero). Yet the dictionary audit of the same adapter is profile-dominated (four of five top features at  $\approx 100\%$  profile mass). Capture without a measurable removal effect in the 3B run, capture with a large one in the 8B run: the difference is consistent with the capacity account of Section 9, but the design does not isolate capacity. Qwen2.5-3B and Qwen3-8B differ in model generation and post-training as well as size, each arm has one adapter seed (the paper’s own 3B r6.2 adapters vary from 0.414 to 0.738 day-level AUC across seeds), and a non-significant 3B effect is not evidence of no effect.

*The 8B collapse is concentrated, and psychometric familiarity tracks it.* The 8B collapse is not uniform across the four malicious users: one remains detected (user-level AUC 0.919), two sit at chance (0.527), and one is *inverted*—ranked 344th of 407 test users, among the least anomalous. With the profile removed, all four are ranked in the top fifth of the cohort at both scales (ranks 10–75 of 407 at 8B, 4–64 at 3B; user-level AUC 0.82–0.99); the one user the 8B FULL adapter still detected moves slightly down ( $0.919 \rightarrow 0.867$ ) and the three collapsed users move up. A direct check of the mechanism against the 3,590 benign training users: the inverted user has the least novel PSY profile of the four (12 benign users within  $L_1 \leq 10$  in the five-dimensional Big Five space) and the surviving user one of the most novel (3); the two most psychometrically familiar profiles belong to the inverted user and to one of the two chance-level users. The most novel profile (0 neighbours) belongs to the other chance-level user, which is however the best detected at day level (0.653), so novelty orders the users only approximately. The organizational header is not predictive (the surviving user shares its role, department, and team with 29 benign users). Ablation and novelty point to the same carrier: the 8B adapter memorizes psychometric familiarity strongly enough that a malicious user whose personality scores resembles many benign trainees is scored as familiar and ranked below the unfamiliar unseen benigns. With four users this is consistent with, not proof of, the mechanism.

*Shuffling.* Deranging profiles never helps. At 8B the SHUFFLE adapter sits with FULL at chance (0.485); at 3B it falls well below the honest-profile adapter (0.656 versus 0.926). Because the derangement preserves per-user constancy, what is lost is not a stable per-user token but the correspondence between the profile’s content and its owner; and because the permutation runs over the whole population, an unseen user can carry a training user’s profile, so profile familiarity no longer tracks whether a user was seen. A fixed profile that belongs to someone else is worse than none, even for a model for which the honest profile is harmless.

*Capture survives shuffling; usefulness does not.* Running the dictionary audit on the FULL and SHUFFLE adapters separately dissociates what the model represents from what helps it. Profile capture does not disappear when profiles are deranged: at 3B the top five features are profile-bound under both adapters (four of five under FULL, five of five under SHUFFLE), and at 8B both adapters place two features of essentially pure profile mass among their top five. The derangement keeps each user’s borrowed profile constant across that user’s days, so the per-user constancy that makes profile tokens cheap to predict is preserved; only the original correspondence between profile content and the user’s behavior and labels is broken; the reassigned profile remains a stable pseudonym for the user. Capture tracks per-user constancy, which shuffling leaves intact, while usefulness tracks whether the profile content is informative about the unseen users being ranked, which shuffling degrades: under SHUFFLE the audit reports the same profile capture alongside chance-level ranking of unseen users. This is a controlled instance of the methodological point that runs through the paper: attribution alone overstates a shortcut, and only a behavioral or causal probe distinguishes an informative binding from a merely salient one.

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## 11 Conclusions

The overall picture can be stated as a two-factor account. The composition of a dataset’s malicious population determines *when* an identity shortcut is available: profile information separates the classes when the malicious users are a small, distinctive minority (CERT r6.2) and separates nothing when they are not (TWOS). The training objective and input representation determine *which* detectors take an available shortcut: the likelihood-based sequence model took it, and three classical feature-vector detectors did not. Both factors were tested rather than assumed. The first by subsampling, by an out-of-setting prediction that the real-user dataset confirmed, by a train-matched retraining experiment showing that profile content in the serialization causes the unseen-user collapse in the audited 8B run (Section 10), and by a login log with no profile at all, where the shortcut relocates to the host field (Section 8); the second by the cross-detector comparison. The retraining experiment adds a qualification the account did not anticipate: the shortcut was captured in both audited runs but measurably damaged unseen-user ranking only in the larger one, a model-dependent difference that the single-seed, cross-family design does not attribute to capacity alone.

Three practical conclusions carry beyond these datasets and this model. First, for anomaly detectors intended to detect behavior, ranking quality alone is not evidence of success; evaluation should compare malicious observations only against normal observations from equally unseen users, and, where possible, within users. Second, including static descriptors of people (role, department, personality measures) in a detector’s input is not harmless; here it created the shortcut in one detector and degraded performance even where the detector was behavioral. Third, the question “what does this detector actually detect?” is answerable: differences of internal states, a sparse dictionary approximation, causal testing of selected directions against matched controls, and attribution of those directions to input content jointly turn the question into a measurable quantity, and the robustness program of Section 6 is what entitles the answer to be believed.

The claims are scoped by design. The causal results concern the particular adaptation runs audited here; the r6.2 estimates rest on four malicious users dominated by one; the CERT data are synthetic; and the real-data studies are small; the retraining experiment uses one adapter seed per scale, its intervals coming from resampling the four malicious users, which characterizes evaluation uncertainty for those adapters and not variation across training runs; and the login-log study is a single 3B adapter per condition with a red-team rather than insider ground truth. What survives every variation we tested is the central dissociation and its explanation, and the method by which both were found.